MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5719

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, subminiature, high Mu

Outline --- 3-1 (EIA)

Base --- E8-10

Envelope --- T3

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	g	nc	h	nc	k	h	nc	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ib	Pp	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	W	°C	ft
Maximum:	6.6	165	0, -55	200	---	1.2	3.3	0.55	+220	note 1
Minimum:	6.0	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	0	1,500	---	---	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1256	Electrode current (1) (anode)		0.4	II	Ib	0.50	0.90	mAdc
1256	Electrode current (2) (anode)	Ec = -2.5 Vdc	0.4	II	Ib	---	50	µAdc
1266	Total grid current	Eb = 150 Vdc; Rk = 2,700 ohms; Rg = 1.0 Meg (see note 2)	0.4	II	Ic	0	-0.3	µAdc
1301	Heater current		0.4	II	If	140	160	mA
1306	Transconductance (1)		0.4	II	Sm	1,400	2,000	µmhos
1336	Heater-cathode leakage		0.4	II	Ihk	---	5.0	µAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		2.5	I	---	---	---	---
1031	Low-frequency vibration	15 G; Rp = 10,000 ohms; F = 40 Hz	2.5	I	Ep	---	25	mVac
1246	Audio frequency noise	Eca = 45 mVac; Rg = 1.0 Meg; Rp = 0.2 Meg (see note 9)	2.5	I	---	---	---	---
1256	Electrode current (3) (anode)	Ec = -1.8 Vdc; Rk = 0	2.5	I	Ib	5	---	µAdc
1266	Grid emission	Ef = 7.5 V; Ec = -2.5 Vdc; Rg = 1.0 Meg Rk = 0 (see note 3)	2.5	I	Ic	0	-0.3	µAdc
1306	Transconductance (2)	Ef = 5.7 V	2.5	I	ΔSm Ef	---	10	%
1316	Amplification factor		---	---	Mu	60	80	---
1321	AC amplification	Ebb = 100 Vdc; Ecc = 0; Esig = 0.2 Vac; Rk = 0	---	---	Ep	8.0	---	Vac
1331	Direct-interelectrode capacitance	No shield	6.5	Code F	Cgp Cin Cout	0.6 1.2 0.4	1.0 2.2 0.8	pF pF pF
1116	Lead fatigue		2.5	Code F	---	---	---	---
2126	Envelope strain		6.5	I	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 2 -Continued							
1041	Shock	450 G; Ehk = + 100 Vdc (see note 4)	---	---	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	See note 5	---	---	---	---
---	Post-shock and vibration-fatigue test end points:							
1031	Low-frequency vibration		---	---	Ep	---	100	mVac
1036	Heater-cathode leakage		---	---	Ihk	---	15	µAdc
1306	Change in transconductance (1) of individual tubes		---	---	ΔSm	---	15	%
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1 ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 7.0 V; 1 min "on", 4 min "off" Ehk = 140 Vac; Eb = Ec = 0	2.5	H	---	---	---	---	---	---
1516	Stability life	Eb = 150 Vdc; Ehk = +200 Vdc; Rg = 1.0 Meg; Rk = 680 ohms; TA = room	---	---	---	---	---	---	---	---
---	Stability life-test end point:									
1306	Change in transconductance (1) of individual tubes		---	---	---	---	ΔSm t	---	10	%
1501	Intermittent life (room temperature)	Stability life-test, or equivalent conditions; TA = room (see notes 6 and 7)	---	---	---	---	---	---	---	---
---	Intermittent life-test end point (room temperature) (500 hours):									
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
---	Intermittent life-test end point (room temperature) (1,000 hours):									
1210	Insulation of electrodes		---	---	1	3	R	25	---	Meg
1501	Intermittent life	Stability life-test conditions; TE = +220°C (min) (see note 8)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current		---	---	1	3	Ic	0	-0.6	µAdc
1301	Heater current		---	---	1	3	If	138	164	mA
1306	Change in trans-conductance (1) of individual tubes		---	---	1	3	ΔSmt	---	20	%
1306	Transconductance (2)		---	---	1	3	ΔSmEf	---	15	%
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	10	µAdc
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
1306	Transconductance (1) average change		---	---	---	---	Avg ΔSmt	---	15	%
---	Total defectives		---	---	3	6	---	---	---	---
---	Intermittent life-test end points (1,000 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current		---	---	1	3	Ic	0	-0.6	µAdc
1301	Heater current		---	---	1	3	If	138	164	mA
1306	Change in trans-conductance (1) of individual tubes		---	---	1	3	ΔSmt	---	25	%
1306	Transconductance (2)		---	---	1	3	ΔSmEf	---	20	%
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	10	µAdc
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
---	Total defectives		---	---	4	8	---	---	---	---

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last performed on the sample selected for grid-emission test.

Ef	Ec	Eb	Rk	Rg
V	Vdc	Vdc	Ohms	Meg
7.5	0	100	1,500	1.0

4. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
6. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.
7. Eligibility for early release at 500 hours is established if no lot failure due to this life test or the regular 1,000-hour intermittent life test has occurred in the preceding 3 consecutive lots and the lot under consideration is acceptable at 500 hours. One lot failing the 1,000-hour intermittent life test or the 1,000-hour intermittent (room temperature) life test, shall result in loss of eligibility for the early release procedure. All intermittent (room temperature) life-test samples shall be operated to 1,000 hours.
8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
9. The rejection level shall be set at the vu meter reading obtained during calibration.
10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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